

NO UNIFORM FINITE INTERPOLATION CONSTANT FOR ALL BANACH PAIRS

RUN FA_BANACH_001

Status. `literature_implied_answer` (negative answer). This note records that a standard later Lipschitz-extension bound negatively answers the finite interpolation question asked by van Neerven–Veraar. It is not a new counterexample from the run.

SOURCE QUESTION

Van Neerven and Veraar’s paper *On the action of Lipschitz functions on vector-valued random sums*, arXiv:math/0504452, asks whether for every pair of Banach spaces X, Y there is a constant $c(X, Y)$, independent of n , such that for all distinct $x_1, \dots, x_n \in X$ and all $y_1, \dots, y_n \in Y$, there is a Lipschitz map $f : X \rightarrow Y$ satisfying $f(x_j) = y_j$ and

$$\|f\|_{\text{Lip}} \leq c(X, Y) \max_{\substack{1 \leq j, k \leq n \\ j \neq k}} \frac{\|y_j - y_k\|}{\|x_j - x_k\|}.$$

They emphasize that the important point is that $c(X, Y)$ should not depend on n , while Johnson–Lindenstrauss–Schechtman give logarithmic constants for each fixed n .

SUPPORTING THEOREM

For metric spaces M, Z and $S \subset M$, let $e(M, S, Z)$ denote the least extension constant for Lipschitz maps $S \rightarrow Z$, and let $e_n(M, Z)$ be the supremum of $e(M, S, Z)$ over all $S \subset M$ with $|S| \leq n$.

Naor–Rabani, *On Lipschitz extension from finite subsets*, arXiv:1506.04398, Section 1.4, equation (1.17), record the Johnson–Lindenstrauss bounds

$$\frac{\sqrt{\log n}}{\sqrt{\log \log n}} \lesssim e_n(\ell_\infty, \ell_2) \lesssim \sqrt{\log n}.$$

In particular $e_n(\ell_\infty, \ell_2) \rightarrow \infty$.

Literature-implied negative answer. *The van Neerven–Veraar interpolation question has a negative answer. Already the fixed pair $X = \ell_\infty$ and $Y = \ell_2$ admits no constant $c(X, Y)$ independent of n .*

Proof. Assume that such a constant $c(\ell_\infty, \ell_2)$ existed. Let $S = \{x_1, \dots, x_m\} \subset \ell_\infty$, with $m \leq n$, and let $g : S \rightarrow \ell_2$ be any Lipschitz map, writing $y_j = g(x_j)$. The source question’s asserted interpolation property would give an extension $f : \ell_\infty \rightarrow \ell_2$ with

$$\|f\|_{\text{Lip}} \leq c(\ell_\infty, \ell_2) \|g\|_{\text{Lip}}.$$

Thus $e(\ell_\infty, S, \ell_2) \leq c(\ell_\infty, \ell_2)$ for every finite S , and hence

$$e_n(\ell_\infty, \ell_2) \leq c(\ell_\infty, \ell_2) \quad \text{for all } n.$$

This contradicts the Johnson–Lindenstrauss lower bound quoted by Naor–Rabani, which implies

$$e_n(\ell_\infty, \ell_2) \gtrsim \frac{\sqrt{\log n}}{\sqrt{\log \log n}} \rightarrow \infty.$$

Therefore no such pairwise constant exists. $\square$

SCOPE AND PROVENANCE

This is a full negative answer to the all-pairs question as stated, by the single pair (ℓ_∞, ℓ_2) . The supporting paper does not explicitly identify arXiv:math/0504452 as the question being answered; the connection is an agent-identified implication through the standard extension constant $e_n(X, Y)$.

The run indexes were searched for arXiv:0504452, the source title, $c(X, Y)$, “distinct elements x_1 ”, “Lipschitz extension”, $e_n(\ell_\infty, \ell_2)$, Johnson–Lindenstrauss, and *On Lipschitz extension from finite subsets*. No exact prior packet for this source problem was found. Existing nearby packets for arXiv:1506.04398 concern different later growth-rate questions.

REFERENCES

- [1] J. van Neerven and M. Veraar, *On the action of Lipschitz functions on vector-valued random sums*, arXiv:math/0504452.
- [2] W. B. Johnson and J. Lindenstrauss, *Extensions of Lipschitz mappings into a Hilbert space*, in *Conference in modern analysis and probability*, Contemp. Math. 26, Amer. Math. Soc., 1984, 189–206.
- [3] W. B. Johnson, J. Lindenstrauss, and G. Schechtman, *Extensions of Lipschitz maps into Banach spaces*, Israel J. Math. 54 (1986), 129–138.
- [4] A. Naor and Y. Rabani, *On Lipschitz extension from finite subsets*, Israel J. Math. 219 (2017), 115–161; arXiv:1506.04398.